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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 04 – Lecture 03: Clustering, Reinforcement Learning & Semi-Supervised ML

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Topic focus: **unsupervised** and **interaction-based** learning paradigms

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning* (2e), selected chapters.;Scikit-learn documentation, accessed 2026-02-28.

Quick Links

Clustering k-means Semi-Supervised RL Basics Summary

Agenda

Clustering

k-means

Semi-supervised learning

Reinforcement Learning

Summary

Learning Outcomes

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- Explain RL terminology (state, action, reward, policy) at a high level.
- Describe semi-supervised learning and when it is helpful.

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- $Q(s, a)$: action-value (Q-function)
- Semi-supervised: labeled set L, unlabeled set U, pseudo-labels

Three Paradigms, Three Feedback Sources

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- **Clustering**: no labels, no rewards; discover groups already present in the data.
- **Semi-supervised**: a few labels + many unlabeled examples; extend a supervised model carefully.
- **RL**: no fixed labels per row; the agent learns from delayed rewards during interaction.

What is Clustering?

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Group data points into clusters such that points in the same cluster are more similar.

Examples:

- customer segmentation

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Examples:

- customer segmentation
- document grouping
- anomaly detection (as "far from clusters")

k-means (Algorithm Sketch)

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3. Update each center as the mean of its assigned points.
4. Repeat until convergence.

k-means Objective

k-means minimizes within-cluster squared distance:

$$\sum_{i=1}^{n} \left\| x_i - \mu_{c(i)} \right\|^2$$

where $\mu_{c(i)}$ is the center of the cluster assigned to x_i.

Mini Worked Example (k-means intuition)

Points in 1D: $[1, 2, 8, 9]$, choose $k = 2$.

- Initialize centers: $\mu_1 = 2$, $\mu_2 = 9$

Repeat until assignments stop changing.

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Mini Worked Example (k-means intuition)

Points in 1D: $[1, 2, 8, 9]$, choose $k = 2$.

- Initialize centers: $\mu_1 = 2$, $\mu_2 = 9$
- Assign: $\{1, 2\} \to \mu_1$ and $\{8, 9\} \to \mu_2$
- Update: $\mu_1 = \text{mean}(1, 2) = 1.5$, $\mu_2 = \text{mean}(8, 9) = 8.5$

Repeat until assignments stop changing.

Choosing k (Practical)

- Elbow method (plot objective vs k)

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- Domain knowledge
- Stability across runs (k-means depends on initialization)

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- Internal metrics (no labels): inertia (k-means objective), silhouette score
- Visual check (2D/3D after PCA) can reveal separation
- Beware: "best" clustering depends on the business goal

Semi-Supervised Learning (Idea)

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Risks: error reinforcement if pseudo-labels are wrong.

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Semi-Supervised Learning (Idea)

- Use a small labeled dataset + large unlabeled dataset.
- Useful when labeling is expensive but unlabeled data is abundant.
- Example approach: **self-training** (pseudo-label confident predictions).

Risks: error reinforcement if pseudo-labels are wrong.

Semi-Supervised Methods (Examples)

- **Self-training**: add confident pseudo-labels iteratively

Use validation to avoid reinforcing mistakes.

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Semi-Supervised Methods (Examples)

- **Self-training**: add confident pseudo-labels iteratively
- **Co-training**: two "views" of features teach each other (when applicable)
- **Graph-based label propagation**: spread labels to neighbors

Use validation to avoid reinforcing mistakes.

Failure Mode: Wrong Pseudo-Labels

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- Start with very few labeled examples from a rare class.
- A weak first model can assign confident but wrong pseudo-labels.
- If we retrain on those errors, the model becomes more biased, not less.

Reinforcement Learning (High Level)

- Agent interacts with environment: **state** $\rightarrow$ **action** $\rightarrow$ **reward**

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- Goal: learn a **policy** that maximizes cumulative reward

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- Goal: learn a **policy** that maximizes cumulative reward
- Feedback is delayed and noisy, so evaluation is harder than in supervised learning

Return and Discounting (Key RL Idea)

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- A discount factor γ controls short-term vs long-term focus.
- High γ values encourage planning for later rewards.

Toy Example: Grid World

- State: agent position on a grid

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Toy Example: Grid World

- State: agent position on a grid
- Actions: up/down/left/right
- Reward: +1 at goal, -1 at trap, small penalty per step
- Learn a policy to reach goal quickly

Summary

- Clustering finds structure without labels; k-means is a common baseline.
- RL learns from reward signals in interaction.
- Semi-supervised learning combines labeled and unlabeled data.

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning* (2e), selected chapters.;Scikit-learn documentation, accessed 2026-02-28.

Exit Questions

1. What is the k-means objective function (in words)?
2. Why can k-means give different clusters on different runs?
3. Define state, action, reward, and policy in RL.
4. Give one benefit and one risk of semi-supervised learning.